

ABHISHEK KUMAR

Principal AI QA Architect & Senior QE Leader | Multi-Agent AI Test Platforms & Enterprise Mobile/Web Automation

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Notice: Immediate / 15–30 Days

EXECUTIVE PROFILE

17+ years of Senior QA Leadership and Enterprise Test Architecture delivering mission-critical quality engineering across Tier-1 FinTech (IG Group — 12 years on FCA/MiFID II-regulated trading platforms), Global Retail (7-Eleven GCC — CoE Architect across 4 geographies), and Frontier AI Quality Consulting (Scale AI). Architect of enterprise multi-agent AI test platforms integrating **LangGraph StateGraph workflows, PostgreSQL pgvector, LangChain chunking, MCP (Model Context Protocol), Playwright, and LangSmith** across web, API, and mobile ecosystems.

GenAI Quality Engineering Authority: Deep hands-on expertise in RAG pipeline evaluation (document chunking boundaries, dense vector retrieval, **Hit Rate @ 3, MRR**), LLM evaluation frameworks (RAGAS, DeepEval), 22-criterion rubric engineering, and prompt-injection threat mitigation (OWASP Top 10 for LLMs). Proven architect of unified automation engines driving native iOS and Android apps via **Appium 2.0 + WebdriverIO 9**, alongside modern responsive web test suites via **Playwright (TypeScript/Java)** and Gatling performance engineering. Slashed test flakiness to <1% and cut regression execution cycles by 85%.

CORE COMPETENCIES

GenAI & Agentic Quality Engineering: LangGraph (StateGraph Multi-Agent Workflows) • PostgreSQL / pgvector (HNSW & IVFFlat Indexing) • LangChain Chunking (RecursiveCharacterTextSplitter) • LLM Evaluation Rubrics • RAGAS • DeepEval • Semantic Search (<=> Cosine Distance) • Prompt Injection & Jailbreak Defense • Hallucination Gating • LangSmith Telemetry • MCP • SHA-256 Prompt Caching

Core Automation Architecture: Playwright (TypeScript/Java) • Page Object Model (POM) • Selenium WebDriver 4 • REST API Automation (APIRequestContext) • GraphQL Testing • Gatling (TypeScript/Scala Load Testing) • JMeter • REST Assured • Cypress • Cucumber BDD

Mobile End-to-End Testing: Appium 2.0 • WebdriverIO 9 • Android (UiAutomator2) • iOS (XCUITest) • Playwright Mobile Emulation • BrowserStack Real-Device Cloud • AWS Device Farm • Deep Linking • Push Notifications • Hybrid App / Webview Testing

Security, Accessibility & Compliance: OWASP API & LLM Top 10 • Row-Level Security (RLS) • OAuth 2.0 / RBAC • WCAG 2.2 AA Accessibility (@axe-core/playwright) • PII Boundary Enforcement • FCA / MiFID II Compliance • PCI-DSS Payment Tokenization

QE Leadership & DevOps: QA CoE Build (0 to 15+ headcount) • Enterprise QE Strategy • Shift-Left & Dual-Speed CI/CD Quality Gates • GitLab CI • GitHub Actions • Jenkins • Docker • AWS • Executive CXO Reporting

PROFESSIONAL EXPERIENCE

AI Quality Engineering Consultant | Scale AI (Independent Advisory) Remote • Apr 2026 – Present

- Frontier Model Evaluation:** Engaged as an independent AI Quality specialist evaluating 4 frontier LLM checkpoints (Claude Opus, GPT-5, Gemini Pro) across multi-turn reasoning, agentic tool invocation, and instruction-following fidelity.
- Rubric Engineering:** Designed 22-criterion binary weighted rubrics (+5/-5 scoring) rigorously evaluating task groundedness, factual accuracy, hallucination boundaries, and privacy violation prevention.
- Adversarial & Guardrail Testing:** Systematically engineered prompt-injection, jailbreak, and role-override test suites to validate safety boundaries and guardrail enforcement against adversarial inputs.
- Vector Semantic Optimization:** Implemented semantic similarity clustering using **PostgreSQL pgvector** (<=> cosine distance) to detect redundant prompt patterns across evaluation batches, accelerating review velocity.
- Top-Tier Performance Rating:** Consistently maintained an exceptional **5.0/5.0 quality rating** across model evaluation accuracy, instruction adherence, and reliability assessments.

Lead Engineer – QA | CoE Architect | 7-Eleven Global Solution Center (GCC)

Bengaluru, India • Dec 2022 – Feb 2026

- **Built Enterprise QA CoE from Scratch:** Scaled and mentored a high-performing 15+ engineer Global QA CoE across 4 geographies (North America, Europe, APAC); owned hiring, quality transformation roadmaps, CI/CD quality gates, and CXO release dashboards.
- **Quality Engineering for "7Bot" (GenAI RAG Product):** Architected automated testing for 7-Eleven's customer-facing conversational RAG assistant (7Bot). Automated document chunking validation for store policy PDFs using **LangChain's RecursiveCharacterTextSplitter** with 15% overlap constraints. Benchmarked **PostgreSQL pgvector** retrieval performance across 128-d dense embeddings, asserting **Hit Rate @ 3 ≥ 75%** and **MRR ≥ 0.65** using native $\cosine$ distance on HNSW indexes. Validated **LangGraph StateGraph workflows** to guarantee deterministic prompt-injection interception at the guardrail layer and graceful fallback routing for out-of-domain queries to prevent hallucinations.
- **Enterprise Mobile Automation (7Now App):** Architected end-to-end mobile automation using **Appium 2.0 + WebdriverIO 9** across Android (UiAutomator2) and iOS (XCUITest), covering cart, payment, dynamic promotions, and order tracking on 20+ real devices via BrowserStack. Automated location-aware mobile scenarios validating GPS-driven store selection and inventory availability.
- **Architected Self-Healing Automation (The Healer Agent):** Engineered an AI-assisted diagnostic locator engine: on UI test failure, it chunks massive 50,000-line DOM snapshots using **LangChain**, retrieves historical locator patterns from **pgvector**, and runs a **LangGraph sandbox loop** to propose verified PR diffs for human review—slashing test maintenance by 40% and cutting flakiness to <1%.
- **Full-Pyramid Quality & Performance:** Automated Store Operations Scorecards and BI Analytics dashboards against underlying REST APIs and SQL databases with 100% data fidelity. Engineered automated Gatling load tests simulating 1,000-user spikes (15s ramp) to enforce CI/CD performance gates (<2s response threshold). Enforced WCAG 2.2 AA accessibility gates using @axe-core/playwright across all digital checkout flows. Optimized GitHub Actions pipelines with cached dependency layers, cutting regression run times from **45 minutes to 6 minutes (85% reduction)**.

QA Automation Developer | Automation QA Lead | IG InfoTech India (IG Group)

Bengaluru, India • Sep 2010 – Dec 2022 • 12 Years

- **Owned Trading Platform Quality (FCA-Regulated):** Led UI and API automation for multi-currency equities, FX, and financial derivatives platforms, validating TCF (Treating Customers Fairly), MiFID II surveillance, and immutable audit trails.
- **Mobile Trading App Automation:** Delivered native iOS & Android trading app test automation (Appium, XCUITest, Espresso) achieving 85%+ automated coverage across critical execution and position-management journeys.
- **High-Throughput Financial API Automation:** Engineered resilient API automation suites covering payment gateways, OAuth 2.0 authentication, and financial transaction settlement.
- **Global Release Coordination:** Orchestrated continuous performance (Gatling/JMeter) and accessibility testing across London, Krakow, and Bengaluru with zero SLA breaches across 12 years.

Programmer Analyst | Infosys

Bengaluru, India • May 2009 – Aug 2010

- Contributed to enterprise eCommerce platform modernization, automating ATG-based transactional workflows.

Associate Technology L2 | Sapient Corporation

Bengaluru, India • Nov 2007 – Jan 2009

- Validated UI and backend transaction engines for scalable telecom eCommerce portals.

KEY REPOSITORY SHOWCASE

Enterprise Multi-Agent AI Test Platform: github.com/abhirise1981/ai-test-automation-platform

Demonstrates a 5-stage multi-agent pipeline (Planner → Generator → Healer) built with Playwright, Appium 2.0, LangGraph, and PostgreSQL pgvector. Includes 35 automated specs testing chunking boundaries, pgvector cosine search ($\cosine$), and LangGraph state machine routing.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Lean Six Sigma Black Belt — GAQM (Lic: G1023899)

ISTQB Certified Tester AI (CT-AI) — BCS/ISEB (Lic: sr9094246)

PSM I – Professional Scrum Master — Scrum.org (Lic: 1028609)

Oracle Certified Professional Java SE 8 — Oracle (Lic: OC1685736)

Executive IT Leadership & Quality Mgmt — XIME Bangalore

AI Tools Mastery Workshop — Be10x

EDUCATION

B.E. in Computer Science — Visvesvaraya Technological University (VTU), Belgaum

Karnataka, India